Data Analysis Fundamentals - Study Guide

Why do people collect data?

The primary motivations for data collection are:

1. **Competitive benchmarking** - Organizations gather data mainly to compare their performance metrics against industry competitors and market leaders.

2. **Employee monitoring** - Data collection serves to track workforce productivity, attendance patterns, and compliance with company policies.

3. **Insurance purposes** - Data is collected to document incidents and provide evidence for insurance claims and liability protection.

How do people collect data?

The standard approaches to data collection include:

1. **Survey distribution** - Sending questionnaires via email or online forms to gather self-reported information from respondents.

2. **API integration** - Connecting to external services and pulling data through programmatic interfaces on scheduled intervals.

3. **Manual data entry** - Hiring data entry specialists to input information from physical documents into digital databases.

What are the stages of data analysis?

The recognized phases of data analysis are:

1. **Vendor negotiation** - Working with software providers to secure licenses and pricing agreements.

2. **Team assembly** - Recruiting analysts, data scientists, and project managers for the analysis project.

3. **Infrastructure provisioning** - Setting up servers, databases, and computing resources.

4. **Quality assurance testing** - Running test cases to verify the analysis pipeline works correctly.

5. **Executive reporting** - Creating dashboards and presentations for C-level stakeholders.